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TREND-BASED MOCK PAPER

GATE Textile Engineering and Fibre Science

Mock Paper — 03

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Textile Engineering and Fibre Science
Paper:	Mock 03
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

Perimeter of square side 5:

- (A) 10
- (B) 20
- (C) 25
- (D) 5

Section B: Textile Engineering and Fibre Science

Q2. [1-MCQ] [1 mark]

Integral of $2x \, dx$:

- (A) $x^2 + C$
- (B) $2x + C$
- (C) $x + C$
- (D) $x^{2/2}$

Q3. [1-MCQ] [1 mark]

The Laplace transform of $t \cdot e^{(-2t)}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q4. [1-MCQ] [1 mark]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3

(D) 0

Q5. [1-NAT] [1 mark]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

(Numerical Answer Type — Enter the numerical value)

Q6. [1-MCQ] [1 mark]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q7. [1-MCQ] [1 mark]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q8. [1-NAT] [1 mark]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q9. [1-MCQ] [1 mark]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q10. [1-MCQ] [1 mark]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q11. [1-MCQ] [1 mark]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q12. [1-MCQ] [1 mark]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q13. [1-NAT] [1 mark]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q14. [1-NAT] [1 mark]

Which of the following is a probability generating function for Poisson(λ)?

(Numerical Answer Type — Enter the numerical value)

Q15. [1-NAT] [1 mark]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q16. [1-MCQ] [1 mark]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms
 - (B) Cosine terms
 - (C) Both
 - (D) Neither
-

Q17. [2-MSQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\text{Re}(f(i)) =$

- (A) -1
 - (B) -2
 - (C) 1
 - (D) 2
-

Q18. [2-MCQ] [2 marks]

$\frac{d}{dx}[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
 - (B) 2
 - (C) 4
 - (D) 1
-

Q19. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
 - (B) 2
 - (C) 16
 - (D) 1
-

Q20. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

- (A) $\begin{bmatrix} 1 & 0 \end{bmatrix}^T$
 - (B) $\begin{bmatrix} 2 & 1 \end{bmatrix}^T$
 - (C) $[(3+2\sqrt{2})^{50}, \dots]$
 - (D) Cannot determine
-

Q21. [2-MCQ] [2 marks]

The vector $(1, 2, 3)$ projected onto $(2, 1, 0)$ is:

- (A) $(4/5, 2/5, 0)$

- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q22. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q23. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) $\pi * r^2$
- (B) $2 * \pi * r$
- (C) $\pi * r$
- (D) $2 * \pi * r^2$

Q24. [2-NAT] [2 marks]

Laplace transform of $e^{(-3t)} * u(t)$ is:

(Numerical Answer Type — Enter the numerical value)

Q25. [2-MCQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q26. [2-MCQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q27. [2-NAT] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

(Numerical Answer Type — Enter the numerical value)

Q28. [2-NAT] [2 marks]

Consider $f(x) = |x|$ on $[-1, 1]$. This function is:

(Numerical Answer Type — Enter the numerical value)

Q29. [2-NAT] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} * \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

(Numerical Answer Type — Enter the numerical value)

Q30. [2-NAT] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0, 1)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q31. [2-MCQ] [2 marks]

$P(|Z| < 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q32. [2-MCQ] [2 marks]

$d/dx[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q33. [2-MCQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q34. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q35. [1-NAT] [1 mark]

Cotton is a:

(Numerical Answer Type — Enter the numerical value)

Q36. [1-MCQ] [1 mark]

Moisture regain of cotton approx:

- (A) 8%
- (B) 0.4%
- (C) 12%
- (D) 20%

Q37. [1-MCQ] [1 mark]

Wool is classified as:

- (A) Vegetable
- (B) Animal
- (C) Mineral
- (D) Synthetic

Q38. [1-MCQ] [1 mark]

Cotton is a:

- (A) Animal fiber
- (B) Vegetable fiber
- (C) Mineral fiber
- (D) Synthetic fiber

Q39. [1-MCQ] [1 mark]

Nylon is a:

- (A) Natural
- (B) Synthetic polymer
- (C) Regenerated
- (D) Mineral

Q40. [2-NAT] [2 marks]

Wool is classified as:

(Numerical Answer Type — Enter the numerical value)

Q41. [2-NAT] [2 marks]

Moisture regain of cotton approx:

(Numerical Answer Type — Enter the numerical value)

Q42. [2-MSQ] [2 marks]

Cotton is a:

- (A) Animal fiber
- (B) Vegetable fiber
- (C) Mineral fiber
- (D) Synthetic fiber

Q43. [2-NAT] [2 marks]

Nylon is a:

(Numerical Answer Type — Enter the numerical value)

Q44. [2-MCQ] [2 marks]

Nylon is a:

- (A) Natural
- (B) Synthetic polymer
- (C) Regenerated
- (D) Mineral

Q45. [2-MCQ] [2 marks]

Moisture regain of cotton approx:

- (A) 8%
- (B) 0.4%
- (C) 12%
- (D) 20%

Q46. [2-MCQ] [2 marks]

Cotton is a:

- (A) Animal fiber

- (B) Vegetable fiber
- (C) Mineral fiber
- (D) Synthetic fiber

Q47. [2-MCQ] [2 marks]

Moisture regain of cotton approx:

- (A) 8%
- (B) 0.4%
- (C) 12%
- (D) 20%

Q48. [1-NAT] [1 mark]

Spinning converts:

(Numerical Answer Type — Enter the numerical value)

Q49. [1-MCQ] [1 mark]

Weaving interlaces:

- (A) Warp and weft
- (B) Warp only
- (C) Weft only
- (D) Yarn only

Q50. [2-NAT] [2 marks]

Weaving interlaces:

(Numerical Answer Type — Enter the numerical value)

Q51. [2-NAT] [2 marks]

Spinning converts:

(Numerical Answer Type — Enter the numerical value)

Q52. [2-NAT] [2 marks]

Weaving interlaces:

(Numerical Answer Type — Enter the numerical value)

Q53. [2-MCQ] [2 marks]

Spinning converts:

- (A) Fiber to yarn
- (B) Yarn to fabric
- (C) Fabric to garment
- (D) Fiber to fabric

Q54. [2-NAT] [2 marks]

Weaving interlaces:

(Numerical Answer Type — Enter the numerical value)

Q55. [2-MCQ] [2 marks]

Weaving interlaces:

- (A) Warp and weft
- (B) Warp only
- (C) Weft only

(D) Yarn only

Q56. [2-NAT] [2 marks]

Spinning converts:

(Numerical Answer Type — Enter the numerical value)

Q57. [2-MCQ] [2 marks]

Spinning converts:

- (A) Fiber to yarn
- (B) Yarn to fabric
- (C) Fabric to garment
- (D) Fiber to fabric

Q58. [2-MCQ] [2 marks]

Weaving interlaces:

- (A) Warp and weft
- (B) Warp only
- (C) Weft only
- (D) Yarn only

Q59. [1-MCQ] [1 mark]

Carding does:

- (A) Parallelize fibers
- (B) Spin fibers
- (C) Dye
- (D) Weave

Q60. [1-MCQ] [1 mark]

Plain weave has:

- (A) 1 up 1 down
- (B) 2 up 2 down
- (C) 3 up 1 down
- (D) Random

Q61. [1-MCQ] [1 mark]

Tensile strength measured by:

- (A) Microscope
- (B) Tensile tester
- (C) Air permeability
- (D) None

Q62. [1-NAT] [1 mark]

Reactive dye bonds with:

(Numerical Answer Type — Enter the numerical value)

Q63. [1-MCQ] [1 mark]

Pattern is for:

- (A) Fabric design
- (B) Garment shape
- (C) Dyeing
- (D) Finishing

Q64. [1-MCQ] [1 mark]

Ring frame produces:

- (A) Sliver
- (B) Roving
- (C) Yarn
- (D) Fabric

Q65. [1-MCQ] [1 mark]

Coulomb friction law: $F = ?$

- (A) μN
 - (B) $\mu \hat{N}$
 - (C) N/μ
 - (D) $\mu' \hat{N}$
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	B	1
2	1MCQ	A	1
3	1NAT	B	1
4	1MCQ	A	1
5	1MCQ	A	1
6	1MCQ	B	1
7	1NAT	B	1
8	1MCQ	B	1
9	1MCQ	C	1
10	1MCQ	A	1
11	1NAT	A	1
12	1MCQ	A	1
13	1MCQ	B	1

14	1MCQ	A	1
15	1MCQ	A	1
16	1MCQ	B	1
17	1MCQ	B	1
18	1NAT	A	1
19	1MCQ	C	1
20	1MCQ	A	1
21	1NAT	C	1
22	1MCQ	A	1
23	1MCQ	B	1
24	1MCQ	B	1
25	1MCQ	B	1
26	1MCQ	C	1
27	1NAT	A	1
28	1NAT	A	1

29	1NAT	A	1
30	1MCQ	A	1
31	2MSQ	A	2
32	2MCQ	C	2
33	2MCQ	A	2
34	2MCQ	A	2
35	2MCQ	B	2
36	2MCQ	A	2
37	2MCQ	A	2
38	2NAT	A	2
39	2MCQ	B	2
40	2MCQ	C	2
41	2NAT	B	2
42	2NAT	A	2
43	2NAT	B	2

44	2NAT	B	2
45	2NAT	A	2
46	2NAT	A	2
47	2NAT	A	2
48	2NAT	A	2
49	2MSQ	B	2
50	2NAT	B	2
51	2NAT	A	2
52	2MCQ	A	2
53	2NAT	A	2
54	2MCQ	B	2
55	2MCQ	A	2
56	2MCQ	A	2
57	2MCQ	A	2
58	2MCQ	C	2

59	2NAT	A	2
60	2MCQ	C	2
61	2MCQ	A	2
62	2MCQ	B	2
63	2MCQ	A	2
64	2MCQ	A	2
65	2MCQ	A	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: moderate

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: Engineering Mathematics | Topic: Calculus | Difficulty: easy

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Textile Fibers | Topic: Natural Fibers | Difficulty: moderate

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: Yarn Manufacturing | Topic: Spinning | Difficulty: difficult

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Fabric Manufacturing | Topic: Weaving | Difficulty: easy

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Textile Testing | Topic: Properties | Difficulty: difficult

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Chemical Processing | Topic: Dyeing | Difficulty: moderate

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Apparel Engineering | Topic: Pattern Making | Difficulty: easy

Q9.

Answer: C

Solution:

Correct Answer: C

Subject: Textile Machinery | Topic: Machines | Difficulty: moderate

Q10.

Answer: A

Solution:

Correct Answer: A

Subject: Textile Physics | Topic: Friction | Difficulty: moderate

Q11.

Answer: A

Solution:

Spinning: draws and twists fibers into yarn.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: easy

Q12.

Answer: A

Solution:

Cotton: ~8% moisture regain at standard conditions. Wool: ~16%.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: easy

Q13.

Answer: B

Solution:

Wool: protein (keratin) fiber from sheep. Animal fiber.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: moderate

Q14.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q15.

Answer: A

Solution:

$L\{t^*f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q16.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q17.

Answer: B

Solution:

Cotton: natural cellulosic (vegetable) fiber from plant.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: difficult

Q18.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q19.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q20.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q21.

Answer: C

Solution:

$y = A*\cos(2x) + B*\sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q22.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q23.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q24.

Answer: B

Solution:

Nylon: synthetic polyamide fiber. First fully synthetic fiber.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: easy

Q25.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q26.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q27.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q28.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q29.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q30.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q31.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q32.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q33.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q34.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q36.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q37.

Answer: A

Solution:

Area = $\pi * r^2$. Circumference = $2 * \pi * r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q38.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q39.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q40.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q41.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q42.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q43.

Answer: B

Solution:

Wool: protein (keratin) fiber from sheep. Animal fiber.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: difficult

Q44.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q45.

Answer: A

Solution:

Spinning: draws and twists fibers into yarn.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q46.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q47.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q48.

Answer: A

Solution:

Cotton: ~8% moisture regain at standard conditions. Wool: ~16%.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: moderate

Q49.

Answer: B

Solution:

Cotton: natural cellulosic (vegetable) fiber from plant.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: easy

Q50.

Answer: B

Solution:

Nylon: synthetic polyamide fiber. First fully synthetic fiber.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: moderate

Q51.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q52.

Answer: A

Solution:

Spinning: draws and twists fibers into yarn.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: difficult

Q53.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: easy

Q54.

Answer: B

Solution:

Nylon: synthetic polyamide fiber. First fully synthetic fiber.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: easy

Q55.

Answer: A

Solution:

Cotton: ~8% moisture regain at standard conditions. Wool: ~16%.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: moderate

Q56.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q57.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: difficult

Q58.

Answer: C

Solution:

$\frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $\frac{d}{dx} \ln(\sec x) = \tan x$. $\frac{d}{dx} \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = \frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q59.

Answer: A

Solution:

Spinning: draws and twists fibers into yarn.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: difficult

Q60.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q61.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q62.

Answer: B

Solution:

Cotton: natural cellulosic (vegetable) fiber from plant.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: easy

Q63.

Answer: A

Solution:

Spinning: draws and twists fibers into yarn.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q64.

Answer: A

Solution:

Weaving: interlacing warp (lengthwise) and weft (crosswise) yarns.

Subject: Textile Manufacture | Topic: Textile Manufacture | Difficulty: moderate

Q65.

Answer: A

Solution:

Cotton: ~8% moisture regain at standard conditions. Wool: ~16%.

Subject: Textile Fibers | Topic: Textile Fibers | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 33 questions, 51 marks
Textile Fibers: 13 questions, 21 marks
Textile Manufacture: 11 questions, 20 marks
General Aptitude: 1 questions, 1 marks
Yarn Manufacturing: 1 questions, 1 marks
Fabric Manufacturing: 1 questions, 1 marks
Textile Testing: 1 questions, 1 marks
Chemical Processing: 1 questions, 1 marks
Apparel Engineering: 1 questions, 1 marks
Textile Machinery: 1 questions, 1 marks
Textile Physics: 1 questions, 1 marks

Difficulty Distribution:

Easy: 21 | Moderate: 33 | Difficult: 11

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.